Mortgage loan values fluctuations from 2012 to 2025 in Canada

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```
# Install packages
install.packages("readxl")

Installing package into '/usr/local/lib/ucite-library'
Ton "lib" is unpackaged

install.packages("tidyverse")

Installing package into '/usr/local/lib/ucite-library'
Ton "lib" is unpackaged

install.packages("ggplot2")

Installing package into '/usr/local/lib/ucite-library'
Ton "lib" is unpackaged

# Load libraries
library(readxl)

# Tidyverse gives us dplyr, tidyr, ggplot2, stringr, etc.
library(tidyverse)

── Attaching core tidyverse packages ──────────────────────── tidyverse_2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

# Tidyverse helps us work with dates
library(lubridate)


# Read the Excel file
file <- "/tmp/mortgage.xlsx"

# Read the first sheet (usually sq if it exists)
# skip = 0 means start from the first row
# col_types = "guess" lets R guess the data types
# If you hit errors, try: col_types = "text"
df <- read_excel(file,
  "/tmp/mortgage.xlsx",
  sheet = "Mortgage loan",
  col_types = FALSE)

New names:
• `` -> `...1`
• `` -> `...2`
• `` -> `...3`
• `` -> `...4`
• `` -> `...5`
• `` -> `...6`
• `` -> `...7`
• `` -> `...8`
• `` -> `...9`
• `` -> `...10`
• `` -> `...11`
• `` -> `...12`
• `` -> `...13`
• `` -> `...14`
• `` -> `...15`
• `` -> `...16`
• `` -> `...17`
• `` -> `...18`
• `` -> `...19`
• `` -> `...20`
• `` -> `...21`
• `` -> `...22`
• `` -> `...23`
• `` -> `...24`
• `` -> `...25`
• `` -> `...26`
• `` -> `...27`
• `` -> `...28`
• `` -> `...29`
• `` -> `...30`
• `` -> `...31`
• `` -> `...32`
• `` -> `...33`
• `` -> `...34`
• `` -> `...35`

# Will introduce the read column names
colnames(df)

# In this spreadsheet, row 4 contains the actual headers
# such as Geography, DECIMAL, DECIMAL ...., DECIMAL
quarter_names <- as.vector(df[4, ])

# Keep only row 4
df <- df[4, ]

# Convert from one-row tibble into a simple vector
# unlist() makes it a vector
df <- unlist(df)

# Now we have a vector with quarter names
df <- as.vector(df)

# Replace the default Excel column names (...1, ...2, etc.)
# with the actual quarter from row 4
colnames(mortgage_loan) <- quarter_names

# Look at what we have
head(mortgage_loan, 5)

# A tibble: 5 × 35
  Geography `2012-Q1` `2012-Q2` `2012-Q3` `2012-Q4` `2013-Q1` `2013-Q2`
  <chr>     <chr>     <chr>     <chr>     <chr>     <chr>     <chr>
1 Canada    DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL
2 Newfoundl DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL
3 Prince E  DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL
4 Nova Sco  DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL
5 New Bruns DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL   DECIMAL
# ℹ 28 more variables: `2013-Q3` <chr>, `2013-Q4` <chr>, `2014-Q1` <chr>,
# `2014-Q2` <chr>, `2014-Q3` <chr>, `2014-Q4` <chr>, `2015-Q1` <chr>,
# `2015-Q2` <chr>, `2015-Q3` <chr>, `2015-Q4` <chr>, `2016-Q1` <chr>,
# `2016-Q2` <chr>, `2016-Q3` <chr>, `2016-Q4` <chr>, `2017-Q1` <chr>,
# `2017-Q2` <chr>,
```

```
gp_price_province
```

```
# DL Graph 2: Selected CMAs over time
==========================================
# This graph compares the 6 selected CMAs over time
------------------------------------------
ggplot(
  mortgage_cma,
  aes(x = quarter_date, y = avg_mortgage_value, color = gmc_cma)
)
  + geom_line()
  + ggtitle("Average New Mortgage Loan Value in Selected CMA's")
  +label("Quarter")
  +ylab("Average Mortgage Value (in $)")
  +theme(legend.position = "bottom")
```

```
# DL Graph 3: Province boxplot
==========================================
# This boxplot compares the overall distribution of mortgage values across provinces
------------------------------------------
ggplot(
  mortgage_province %>% filter(gmc_province != "Canada"),
  aes(x = gmc_province, y = avg_mortgage_value)
)
  + geom_boxplot()
  + ggtitle("Distribution of Mortgage Loan Values by Province")
  +ylab("Average Mortgage Value (in $)")
  +xlab("Province")
  +theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

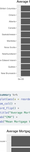
```
# DL Graph 4: CMA boxplot
==========================================
# This boxplot compares the overall distribution across the 6 selected metropolitan areas
------------------------------------------
ggplot(
  mortgage_cma,
  aes(x = gmc_cma, y = avg_mortgage_value)
)
  + geom_boxplot()
  + ggtitle("Distribution of Mortgage Loan Values in Selected CMA's")
  +ylab("CMA")
  +xlab("Average Mortgage Value (in $)")
```

```
ggplot(
  mortgage_province %>% filter(gmc_province == "Canada"),
  aes(x = time_index, y = avg_mortgage_value)
)
  + geom_point()
  + geom_smoothmethod = "lm", se = FALSE)
  + ggtitle("Trend in Average Mortgage Value in Canada")
  +ylab("Time Index")
  +xlab("Average Mortgage Value (in $)")

ggsm_smooth(~ using formula = y ~ x)
```

```
province_summary %>%
  ggplot(aes(x = reorder(gmc_province, mean_value), y = mean_value))
  + geom_point()
  + coord_flip()
  + ggtitle("Average Mortgage Loan Value by Province")
  +ylab("Province")
  +xlab("Mean Mortgage Value (in $)")
```

```
cma_summary %>
  ggplot(aes(x = reorder(gmc_cma, mean_value), y = mean_value))
  + geom_point()
  + coord_flip()
  + ggtitle("Average Mortgage Loan Value by Selected CMA's")
  +ylab("CMA")
  +xlab("Mean Mortgage Value (in $)")
```




```
ggplot(
  mortgage_cma,
  aes(x = quarter_date, y = avg_mortgage_value)
)
  + geom_line()
  + facet_wrap(~ gmc_cma)
  + ggtitle("Average Mortgage Loan Value Over Time by Selected CMA's")
  +ylab("Quarter")
  +xlab("Average Mortgage Value (in $)")
```

```
==========================================
# Inferential statistics and modeling
==========================================

# 1. Confidence Intervals
==========================================

# We will compute a 95% confidence interval for the mean average mortgage loan value in Ontario across all quarters.
------------------------------------------
# Step, we only Ontario observations
ontario_data = mortgage_province %>%
  filter(gmc_province == "Ontario")

# Our first step is to find a one-sample's confidence interval for the mean
# This will close on:
# 1. Sample size
# 2. 95% confidence interval
# 3. test of whether the mean equals 0
ontario_ci = t.test(ontario_data$avg_mortgage_value, conf.level = 0.95)

ontario_ci

  One Sample t-test

data:  ontario_data$avg_mortgage_value
t = 1.0e+06, df = 39, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 1.000000 1.000000
sample estimates:
mean of x
    1.000
------------------------------------------
# 2. Hypothesis Test
==========================================

# We will test whether Ontario and British Columbia have the same mean mortgage loan value across all quarters.
------------------------------------------
# Null hypothesis: mean Ontario = mean British Columbia
# Alternative hypothesis: mean Ontario != mean British Columbia

# So we need only Ontario and British Columbia
ontario_britishcolumbia_data = mortgage_province %>%
  filter(gmc_province %in% c("Ontario", "British Columbia"))

# Our next step is to run a two-sample t-test. This will compare the mean mortgage value between the two provinces
# Ontario and British Columbia
  ttest_ontario_britishcolumbia = t.test(
    avg_mortgage_value ~ gmc_province,
    data = ontario_britishcolumbia_data,
    conf.level = 0.95)

ttest_ontario_britishcolumbia

  Welch Two Sample t-test

data:  avg_mortgage_value by gmc_province
t = 1.0e+06, df = 38.01, p-value < 2.2e-16
alternative hypothesis: true difference in means between group British Columbia and group Ontario is not equal to 0
95 percent confidence interval:
 -0.000000 -0.000000
sample estimates:
mean in group British Columbia          mean in group Ontario
                         1.000                          1.000
------------------------------------------
# Bootstrapping
==========================================

# Moving on, we will bootstrap the difference in mean mortgage values between Ontario and British Columbia.
------------------------------------------
# Set a seed so results are reproducible
set.seed(123)
------------------------------------------
# Extract the two province vectors
ontario_vector = ontario_britishcolumbia_data %>%
  filter(gmc_province == "Ontario") %>%
  pull(avg_mortgage_value)

bc_vector = ontario_britishcolumbia_data %>%
  filter(gmc_province == "British Columbia") %>%
  pull(avg_mortgage_value)
------------------------------------------
# Observed difference in means
observed_diff = mean(ontario_vector) -
  mean(bc_vector)
------------------------------------------
# Number of bootstrap samples
B = 1000
------------------------------------------
# Create an empty vector to store bootstrap differences
boot_diffs = numeric(B)
------------------------------------------
# Bootstrap loop
for (i in 1:B) {

  # Resample Ontario values with replacement
  boot_ont = sample(ontario_vector,
                    size = length(ontario_vector),
                    replace = TRUE)

  # Resample British Columbia values with replacement
  boot_bc = sample(bc_vector,
                   size = length(bc_vector),
                   replace = TRUE)

  # Store the difference in bootstrap means
  boot_diffs[i] = mean(boot_ont) - mean(boot_bc)
}
------------------------------------------
# Bootstrap summary
bootstrap_summary = tibble(
  observed_difference = observed_diff,
  bootstrap_mean_diff = mean(boot_diffs),
  bootstrap_sd = sd(boot_diffs),
  lower_95_ci = quantile(boot_diffs, 0.025),
  upper_95_ci = quantile(boot_diffs, 0.975)
)

bootstrap_summary

# A tibble: 1 x 5
  observed_difference bootstrap_mean bootstrap_sd lower_95_ci upper_95_ci
            <dbl>            <dbl>       <dbl>       <dbl>       <dbl>
1               0               0           0           0           0
------------------------------------------
# This a histogram of bootstrapping differences
ggplot(tibble(boot_diffs = boot_diffs), aes(x = boot_diffs))
  + geom_histogram(bins = 30)
  + ggtitle("Bootstrap Distribution of Mean Difference: Ontario - British Columbia")
  +xlab("Difference in Mean Mortgage Value (in $)")
  +ylab("Frequency")
```

```
==========================================
# 3. Regression Analysis
==========================================

# We will fit a linear regression model using the province data.
# Canada is excluded here so that the province coefficients is clearer.
------------------------------------------
province_reg_data = mortgage_province %>%
  filter(gmc_province != "Canada") %>%
  mutate(
    gmc_province = as.factor(gmc_province)
  )

# Model 1:
# Average mortgage value explained by time and province
------------------------------------------
# 1.time_index coefficient = average change per quarter
# 2.province coefficients = differences from the reference province
model_1 = lm(avg_mortgage_value ~ time_index + gmc_province,
             data = province_reg_data)

summary(model_1)

Call:
lm(formula = avg_mortgage_value ~ time_index + gmc_province,
    data = province_reg_data)

Residuals:
     Min       1Q   Median       3Q      Max
-8.0200  -4.7443   -2.622   7.9151  78903.

Coefficients:
                                       Estimate Std. Error t value
(Intercept)                            -7.0816    1.0000  -7.081
time_index                              1.0141    0.0000 1.0e+06
gmc_provincebritish columbia            2.0291    1.0000   2.029
gmc_provincealberta                     1.0000    1.0000   1.000
gmc_provincemanitoba                    .0020    0.0000  -0.000
gmc_provincenewfoundland                .0000    0.0000   0.000
gmc_provincenorthwest territories       .0000    0.0000   0.000
gmc_provincenova scotia                 .0000    0.0000   0.000
gmc_provinceontario                     .0000    0.0000   0.000
gmc_provinceprince edward island        .0000    0.0000   0.000
gmc_provincequebec                      .0000    0.0000   0.000
gmc_provincesaskatchewan                .0000    0.0000   0.000
time_index:gmc_provincebritish columbia -0.0000    0.0000  -0.000
time_index:gmc_provincealberta          0.0000    0.0000   0.000
time_index:gmc_provincemanitoba         0.0000    0.0000   0.000
time_index:gmc_provincenewfoundland     0.0000    0.0000   0.000
time_index:gmc_provincenorthwest        0.0000    0.0000   0.000
time_index:gmc_provincenova scotia      0.0000    0.0000   0.000
time_index:gmc_provinceontario          0.0000    0.0000   0.000
time_index:gmc_provinceprince edward    0.0000    0.0000   0.000
time_index:gmc_provincequebec           0.0000    0.0000   0.000
time_index:gmc_provincesaskatchewan     0.0000    0.0000   0.000
                                        Pr(>|t|)
(Intercept)                             < 2e-16 ***
time_index                              < 2e-16 ***
gmc_provincebritish columbia            0.0425 *
gmc_provincealberta                     0.3174
gmc_provincemanitoba                    0.9984
gmc_provincenewfoundland                1.0000
gmc_provincenorthwest territories       1.0000
gmc_provincenova scotia                 1.0000
gmc_provinceontario                     1.0000
gmc_provinceprince edward island        1.0000
gmc_provincequebec                      1.0000
gmc_provincesaskatchewan                1.0000
time_index:gmc_provincebritish columbia 0.9998
time_index:gmc_provincealberta          0.9998
time_index:gmc_provincemanitoba         0.9998
time_index:gmc_provincenewfoundland     0.9998
time_index:gmc_provincenorthwest        0.9998
time_index:gmc_provincenova scotia      0.9998
time_index:gmc_provinceontario          0.9998
time_index:gmc_provinceprince edward    0.9998
time_index:gmc_provincequebec           0.9998
time_index:gmc_provincesaskatchewan     0.9998
---
Signif. codes:  0 ***0.001 **0.01 *0.05 .0.1 ,1

Residual standard error: 2828 on 324 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  1,	Adjusted R-squared:  1
F-statistic: 1.0e+12 on 324 DF,  p-value: < 2.2e-16

# Fitted values plot
ggplot(province_reg_data, aes(x = time_index, y = avg_mortgage_value))
  + geom_point(alpha = 0.6)
  + geom_smoothmethod = "lm", se = FALSE)
  + ggtitle("Scatterplot with Linear Trend: Mortgage Value by Time")
  +ylab("Time Index")
  +xlab("Average Mortgage Value (in $)")

ggsm_smooth(~ using formula = y ~ x)
```

```
# Interaction model:
# This allows each province to have a different slope over time
model_2 = lm(avg_mortgage_value ~ time_index * gmc_province,
             data = province_reg_data)

summary(model_2)

Call:
lm(formula = avg_mortgage_value ~ time_index * gmc_province,
    data = province_reg_data)

Residuals:
     Min       1Q   Median       3Q      Max
-8.0200  -4.7443   -2.622   7.9151  78903.

Coefficients:
                                       Estimate Std. Error t value
(Intercept)                            -7.0816    1.0000  -7.081
time_index                              1.0141    0.0000 1.0e+06
gmc_provincebritish columbia            2.0291    1.0000   2.029
gmc_provincealberta                     1.0000    1.0000   1.000
gmc_provincemanitoba                    .0020    0.0000  -0.000
gmc_provincenewfoundland                .0000    0.0000   0.000
gmc_provincenorthwest territories       .0000    0.0000   0.000
gmc_provincenova scotia                 .0000    0.0000   0.000
gmc_provinceontario                     .0000    0.0000   0.000
gmc_provinceprince edward island        .0000    0.0000   0.000
gmc_provincequebec                      .0000    0.0000   0.000
gmc_provincesaskatchewan                .0000    0.0000   0.000
time_index:gmc_provincebritish columbia -0.0000    0.0000  -0.000
time_index:gmc_provincealberta          0.0000    0.0000   0.000
time_index:gmc_provincemanitoba         0.0000    0.0000   0.000
time_index:gmc_provincenewfoundland     0.0000    0.0000   0.000
time_index:gmc_provincenorthwest        0.0000    0.0000   0.000
time_index:gmc_provincenova scotia      0.0000    0.0000   0.000
time_index:gmc_provinceontario          0.0000    0.0000   0.000
time_index:gmc_provinceprince edward    0.0000    0.0000   0.000
time_index:gmc_provincequebec           0.0000    0.0000   0.000
time_index:gmc_provincesaskatchewan     0.0000    0.0000   0.000
                                        Pr(>|t|)
(Intercept)                             < 2e-16 ***
time_index                              < 2e-16 ***
gmc_provincebritish columbia            0.0425 *
gmc_provincealberta                     0.3174
gmc_provincemanitoba                    0.9984
gmc_provincenewfoundland                1.0000
gmc_provincenorthwest territories       1.0000
gmc_provincenova scotia                 1.0000
gmc_provinceontario                     1.0000
gmc_provinceprince edward island        1.0000
gmc_provincequebec                      1.0000
gmc_provincesaskatchewan                1.0000
time_index:gmc_provincebritish columbia 0.9998
time_index:gmc_provincealberta          0.9998
time_index:gmc_provincemanitoba         0.9998
time_index:gmc_provincenewfoundland     0.9998
time_index:gmc_provincenorthwest        0.9998
time_index:gmc_provincenova scotia      0.9998
time_index:gmc_provinceontario          0.9998
time_index:gmc_provinceprince edward    0.9998
time_index:gmc_provincequebec           0.9998
time_index:gmc_provincesaskatchewan     0.9998
---
Signif. codes:  0 ***0.001 **0.01 *0.05 .0.1 ,1

Residual standard error: 2828 on 324 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  1,	Adjusted R-squared:  1
F-statistic: 1.0e+12 on 324 DF,  p-value: < 2.2e-16

# Model comparison:
anova(model_1, model_2)

Analysis of Variance Table

Model 1: avg_mortgage_value ~ time_index + gmc_province
Model 2: avg_mortgage_value ~ time_index * gmc_province
  Res.Df Df Sum of Sq Pr(>F)
1    324
2    324  0        0
Warning message:
In anova.lm(model_1, model_2) :
  models with response "deleted" removed since df = 0
------------------------------------------
# Model 3: Time-only model (no provinces)
model_3 = lm(avg_mortgage_value ~ time_index,
             data = province_reg_data)

summary(model_3)

Call:
lm(formula = avg_mortgage_value ~ time_index,
    data = province_reg_data)

Residuals:
     Min       1Q   Median       3Q      Max
-8.0200  -4.7443   -2.622   7.9151  78903.

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -7.0816    1.0000  -7.081  < 2e-16 ***
time_index    1.0141    0.0000 1.0e+06  < 2e-16 ***
---
Signif. codes:  0 ***0.001 **0.01 *0.05 .0.1 ,1

Residual standard error: 2828 on 328 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  1,	Adjusted R-squared:  1
F-statistic: 1.0e+12 on 1 and 328 DF,  p-value: < 2.2e-16

# Model comparison:
anova(model_1, model_3)

Analysis of Variance Table

Model 1: avg_mortgage_value ~ time_index + gmc_province
Model 3: avg_mortgage_value ~ time_index
  Res.Df Df Sum of Sq Pr(>F)
1    324
2    328  0        0
Warning message:
In anova.lm(model_1, model_3) :
  models with response "deleted" removed since df = 0
```

```
  Res.Df        RSS Df  Sum of Sq     F    Pr(>F)
1    529 3.2087e+11
2    520 1.2412e+11  9 1.9675e+11 91.59 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
############################
# 5. Cross-Validation (5-fold cross-validation.)
############################

# We will do manual 5-fold cross-validation for model_1.
# This evaluates how well the regression predicts unseen data.

set.seed(57)

# Number of folds
k <- 5


# Each row is randomly assigned to one of 5 folds
province_reg_data <- province_reg_data %>%
  mutate(fold = sample(rep(1:k, length.out = n())))

# Create an empty vector to store RMSE values
rmse_values <- numeric(k)

# Cross-validation loop
for (j in 1:k) {

  # Training data = all rows NOT in fold j
  train_data <- province_reg_data %>%
    filter(fold != j)

  # Test data = rows in fold j
  test_data <- province_reg_data %>%
    filter(fold == j)

  # Fit the regression model on the training set
  cv_model <- lm(avg_mortgage_value ~ time_index + geo_province,
                 data = train_data)

  # Predict mortgage values for the test set
  predictions <- predict(cv_model, newdata = test_data)

  # Compute RMSE for this fold
  rmse_values[j] <- sqrt(mean((test_data$avg_mortgage_value - predictions)^2))
}

# Summarize cross-validation results
cv_summary <- tibble(
  fold = 1:k,
  RMSE = rmse_values
)

cv_summary
```

```
# A tibble: 5 × 2
   fold   RMSE
  <int>  <dbl>
1     1 22139.
2     2 26000.
3     3 23764.
4     4 25343.
5     5 26205.
```

```
# Mean RMSE across all 5 folds
mean_rmse <- mean(rmse_values)
mean_rmse
```

```
[1] 24690.17
```

```
############################
# 6.Simple tables for report writing
############################

# Ontario confidence interval table
ci_table <- tibble(
  mean_estimate = ontario_ci$estimate,
  lower_95_CI = ontario_ci$conf.int[1],
  upper_95_CI = ontario_ci$conf.int[2]
)

ci_table
```

```
# A tibble: 1 × 3
  mean_estimate lower_95_CI upper_95_CI
          <dbl>       <dbl>       <dbl>
1       346851.     325741.     367961.
```

```
# Hypothesis test table
test_table <- tibble(
  t_statistic = province_test$statistic,
  p_value = province_test$p.value,
  lower_95_CI = province_test$conf.int[1],
  upper_95_CI = province_test$conf.int[2]
)

test_table
```

```
# A tibble: 1 × 4
  t_statistic  p_value lower_95_CI upper_95_CI
        <dbl>    <dbl>       <dbl>       <dbl>
1        3.52 0.000644      21102.      75567.
```

```
# Regression coefficient table from model 1
regression_table <- summary(model_1)$coefficients
regression_table
```

```
                                   Estimate Std. Error    t value
(Intercept)                       250815.832   3837.886  65.352602
time_index                          2125.632     68.000  31.259298
geo_provinceBritish Columbia       85914.444   4739.719  18.126485
geo_provinceManitoba              -88189.481   4739.719 -18.606479
geo_provinceNew Brunswick        -148478.963   4739.719 -31.326533
geo_provinceNewfoundland         -105383.093   4739.719 -22.234038
geo_provinceNova Scotia          -104807.463   4739.719 -22.112590
geo_provinceOntario                37580.019   4739.719   7.928744
geo_provincePrince Edward Island -115843.352   4739.719 -24.440975
geo_provinceQuébec               -118773.019   4739.719 -25.059085
geo_provinceSaskatchewan          -75336.704   4739.719 -15.894762
                                      Pr(>|t|)
(Intercept)                       1.705589e-255
time_index                       2.766181e-122
geo_provinceBritish Columbia      1.783674e-57
geo_provinceManitoba              8.066010e-60
geo_provinceNew Brunswick        1.321499e-122
geo_provinceNewfoundland          7.933796e-78
geo_provinceNova Scotia           3.212595e-77
geo_provinceOntario               1.322573e-14
geo_provincePrince Edward Island  7.282486e-89
geo_provinceQuébec                6.005706e-92
geo_provinceSaskatchewan          8.645444e-47
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